

Tail-Latency-Aware Segmented Waveguide-Enabled Pinching-Antenna Systems for Hyper-Reliable Low-Latency Communications Under Bursty Traffic

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Abstract—Segmented waveguide-enabled pinching-antenna systems (SWANs) replace the single long dielectric waveguide of a pinching-antenna system (PASS) by several short, individually fed segments, each of which activates one pinching antenna (PA). SWANs thereby provide reconfigurable line-of-sight channels without inter-antenna radiation and can serve users either by aggregating all segments or by multiplexing them. Existing SWAN and PASS designs, however, optimize average rates under full-buffer traffic, whereas 6G hyper-reliable low-latency communication (HRLLC) is specified by the *tail* of the packet delay under bursty, deadline-constrained arrivals. This paper studies a downlink multi-user SWAN whose users generate packets according to Markov-modulated ON-OFF processes, transmit with finite-blocklength (FBL) codes, and discard packets whose delay budget expires, and whose segments consume circuit power when active and need a configuration time when activated. We formulate the joint PA placement, segment activation and mode selection, user scheduling, and rate/power adaptation problem that minimizes the energy and configuration cost under per-user delay-violation-probability constraints, and propose a two-timescale tail-latency-aware scheme, TLA-SWAN. On the frame timescale, a closed-form mapping from each user's burst statistics and tail target to a *tail-latency-aware required service rate* drives a low-complexity block-coordinate PA placement that minimizes the tail load, i.e., the time share needed to drain every user at its required rate. On the slot timescale, a packet-value scheduler assigns to every queued packet a value that grows exponentially with its age at a rate set by the user's tail exponent, and minimizes the drift of the resulting tail-risk potential plus an energy and configuration penalty by jointly selecting the served users, the active segments (segment selection, partial or full aggregation, or multiplexing), the number of packets per codeword, and the power level through an exhaustive search over pre-computed tables. Extensive slot-level simulations with FBL decoding errors and deadline-constrained queues show that controllers with similar average delays differ by orders of magnitude in their delay tails, that TLA-SWAN reduces the delay-violation probability substantially with respect to rate-centric SWAN designs, classical deadline-aware schedulers, fixed segment modes, conventional single-waveguide PASS, and fixed antenna arrays, that full segment activation is not energy-optimal once circuit power and configuration delays are accounted for, that bursty arrivals change the optimal segment policy with respect to Poisson arrivals of the same mean, and that a closed-form effective-bandwidth approximation predicts the single-device tail.

Index Terms—Pinching-antenna systems, segmented waveguide, hyper-reliable low-latency communication, tail latency,

bursty traffic, finite blocklength, Lyapunov optimization, scheduling.

I. INTRODUCTION

HYPER-RELIABLE and low-latency communication (HRLLC) is one of the six usage scenarios of IMT-2030, with research targets of 0.1–1 ms air-interface latency and reliability between $1 - 10^{-5}$ and $1 - 10^{-7}$ [1]. Such targets are not statements about the *average* delay: they bound the probability that the delay of a packet exceeds a budget, i.e., the tail of the delay distribution [2]. Two features of HRLLC applications make the tail particularly hard to control. First, the traffic is bursty: industrial closed-loop control, event-triggered alarms, and extended-reality (XR) frames produce packets in clusters separated by silent periods [3]–[5], so that even a lightly loaded link experiences short overload periods during which queues build up. Second, HRLLC packets are short, so that finite-blocklength (FBL) coding penalties and decoding errors must be accounted for jointly with queueing [6]–[9]. The wireless link therefore needs enough *instantaneous* capacity to drain a burst within the delay budget, which in turn calls for physical-layer architectures that can bring high signal-to-noise ratio (SNR) to wherever a burst occurs.

Pinching-antenna systems (PASS) are a recently proposed flexible-antenna technology in which small dielectric particles (pinching antennas, PAs) are attached at adjustable positions along a dielectric waveguide fed by the base station (BS) [10]–[13]. By activating a PA close to a user, PASS shortens the free-space path and establishes a strong line-of-sight (LoS) link, which makes it attractive for millimeter-wave deployments in factories and indoor venues. A single long waveguide, however, suffers from in-waveguide propagation loss and from inter-antenna radiation (IAR) when several PAs are activated. Segmented waveguide-enabled PASS (SWAN) [14] solves both problems by dividing the waveguide into short segments, each with its own feed connected to the BS by fiber or cable, and by activating a single PA per segment. The feeds can be driven by separate radio-frequency (RF) chains, so that a SWAN behaves like a distributed antenna array whose element positions are reconfigurable within their segments. This gives rise to three operating modes [14], [15]: segment selection (one segment serves a user), segment aggregation (SA, all segments coherently serve one user), and segment multiplexing (SM, the segments spatially multiplex several users). Existing SWAN studies characterize the SNR and sum-

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rate of these modes [14], [16], [17], design hybrid beamformers [18], [19], integrate sensing [20]–[22], and design random-access protocols [15]. All of them assume saturated (full-buffer) traffic and infinite-blocklength coding; none considers queueing, deadlines, or the delay tail.

The combination of SWAN and HRLLC raises three questions that the rate-centric literature does not answer: where the PAs should be placed when users differ in burstiness and tail targets, since a sum-rate placement favors users with good channels whereas the tail is governed by the users whose bursts are hardest to drain; how the BS should decide, slot by slot, between segment selection, aggregation, and multiplexing, how many short packets to bundle in a codeword, and at which power, when a decoding failure delays all bundled packets; and how fast the SWAN can be reconfigured, since PA repositioning and segment activation take from microseconds to seconds [11], [23], so that chasing bursts may cost more than it gains.

A. Related Work

PASS and SWAN. The channel and signal models of PASS were established in [10] and are surveyed in [11], [12]. SWANs were introduced in [14], which derived the SNR scaling laws of the segment-selection, aggregation, and multiplexing modes; follow-up work addressed uplink sum-rate maximization [16], joint segment activation and placement under a single RF chain [17], tri-hybrid beamforming [18], sensing [20], and access protocols that exploit the reconfigurable channel diversity of SWANs [15]; timeliness has been considered only through the age of information [24]. For single-waveguide PASS, FBL rate optimization for one user was studied in [25] and a statistical-CSI short-packet design in [26]; both consider one user, one packet, and no queue. The cost of PA repositioning has been modeled through a finite movement speed [23] and discrete activation [27], again without arrivals or deadlines, and a recent network-level perspective [28] lists queue-aware PA scheduling under discrete activation constraints as an open problem, which is precisely the problem addressed here.

Tail latency under bursty traffic and FBL. The delay tail of a wireless queue is classically characterized by effective bandwidth and effective capacity [29]–[32], stochastic network calculus [33], [34], and extreme value theory [2]; Markov-modulated ON–OFF sources are the standard model of bursty arrivals, with closed-form effective bandwidths [35], [36]. In the URLLC context, [37] converted a queueing-delay-violation constraint into a service-rate requirement through the effective bandwidth of the arrivals, [38] incorporated FBL coding into the tail analysis, [8] adapted the blocklength to the queue state, and [9] characterized the reliability–latency–rate trade-off. Tail-aware schedulers include the M-LWDF rule [39], the exponential rule [40], and largest-weighted-delay-first scheduling, which is large-deviations optimal [41]; Lyapunov drift-plus-penalty methods [42], [43] turn delay constraints into virtual queues and have been applied to URLLC edge computing [44] and hard-latency-constrained scheduling [5]. None of these works involves a reconfigurable antenna geometry. Among reconfigurable antennas, fluid antennas have

been studied for HRLLC with an explicit port-switching delay that shortens the effective blocklength [45], and reconfigurable intelligent surfaces for URLLC with two-timescale designs [46]; these works confirm that reconfiguration overhead must be charged against the latency budget but do not model queues or bursty arrivals.

B. Contributions

To the best of our knowledge, this is the first work that studies a SWAN (or any PASS) with packet arrivals, queues, deadlines, and FBL coding, and the first that designs the PA geometry from the delay tail rather than from the rate. The contributions are as follows.

- *Model and formulation.* We model a downlink multi-user SWAN with per-segment RF chains that can be switched on and off, circuit power per active segment, a configuration delay per activation or PA move, LoS-dominant PA channels, Markov-modulated ON–OFF packet arrivals, FBL transmissions with retransmission upon decoding failure, and deadline-based packet discarding. We formulate the joint PA placement, segment activation and mode selection, scheduling, and rate/power adaptation problem that minimizes energy and configuration cost under per-user delay-violation-probability constraints.
- *Tail-latency-aware required rate.* We map the ON–OFF burst parameters, the delay budget $D_{\max}$, and the violation target δ of a user into a single scalar, the tail-latency-aware required service rate c_{req} , in closed form. This scalar summarizes “how fast a burst must be drained”; the tail exponent $\Lambda_k = \ln(1/\delta_k)/D_{\max}$ that defines it is the common currency of both timescales of the proposed scheme.
- *Two-timescale TLA-SWAN scheme.* On the frame timescale, a block-coordinate-descent placement (TAPP) minimizes the tail load, the sum over the users of the ratio between the required rate and the FBL rate obtained from the user’s two best segments, which moves PAs towards bursty and stringent users rather than towards high-rate users. On the slot timescale, a tail-aware scheduler (TAS) values every queued packet exponentially in its age with the user’s tail exponent, so that the sum of the values is a tail-risk potential whose growth measures the imminence of deadline violations, and minimizes the drift of this potential plus the energy and configuration penalty by jointly selecting the users, the active segments (from segment selection to full aggregation, or multiplexing), the packets per codeword, and the power level through an exhaustive search over tables that are computed once per frame; a one-step lookahead lets the controller pay a configuration delay when a burst justifies it. The per-slot complexity is that of a table lookup.
- *Guarantees.* We state the convergence of TAPP, the closed-form single-user placement, the drift-plus-penalty characterization of TAS, and an effective-capacity bound on the delay-violation probability (proofs in the appendices).
- *Evaluation.* A slot-level simulator with FBL decoding errors, deadline dropping, circuit power, and configuration delays tests four hypotheses: that average delay hides tail violations, that full segment activation is not always optimal,

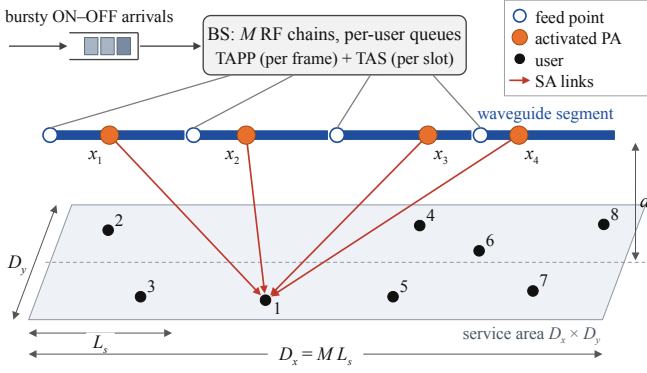


Fig. 1. Downlink SWAN with $M = 4$ segments. Each segment is fed at its left end by its own RF chain and activates one PA at x_m ; the BS chooses per slot between segment selection, segment aggregation (SA, shown for the bursting user 1), and segment multiplexing.

that tail-aware control beats rate- and delay-oriented control at lower energy, and that bursty arrivals change the optimal segment policy. TLA-SWAN is compared with rate-centric placement, max-weight, rate-maximizing, EDF, and M-LWDF control, fixed segment modes, reactive repositioning, conventional PASS, and a fixed array; a single-device case validates the effective-bandwidth tail approximation.

C. Organization and Notation

Sections II–V present the system model, the problem, the proposed scheme, and the results; Section VI concludes. Boldface lowercase and uppercase letters denote vectors and matrices; $(\cdot)^T$ and $(\cdot)^H$ denote transpose and conjugate transpose; $|\cdot|$ is the modulus of a scalar or the cardinality of a set; $\mathbb{1}\{\cdot\}$ is the indicator function; $[x]^+ = \max\{x, 0\}$; $Q(\cdot)$ is the Gaussian Q -function; and $\mathcal{CN}(0, \sigma^2)$ is the circularly symmetric complex Gaussian distribution.

II. SYSTEM MODEL

A. SWAN Architecture

We consider the downlink of a SWAN, illustrated in Fig. 1. Following [14], M dielectric waveguide segments of length L_s are placed end-to-end along the x -axis at height d above the ground plane, so that segment $m \in \mathcal{M} = \{1, \dots, M\}$ spans $[(m-1)L_s, mL_s]$ and the service area has length $D_x = ML_s$. The feed point of segment m is located at its left end, $\psi_m^0 = [(m-1)L_s, 0, d]^T$, and is connected to the BS by an individual RF chain and power amplifier with power budget $P_{\max}$. Because the segments are not physically connected, the signal captured or radiated by one segment cannot leak into another, and activating a single PA per segment avoids IAR [14]. The activated PA of segment m is located at $\psi_m = [x_m, 0, d]^T$ with

$$x_m \in \mathcal{X}_m \triangleq [(m-1)L_s + \epsilon_x, mL_s - \epsilon_x], \quad (1)$$

where ϵ_x is a small guard distance from the segment ends. We write $\mathbf{x} = [x_1, \dots, x_M]^T$.

Time is slotted with slot duration T_s and index $t = 0, 1, \dots$; slots are grouped into frames of T_f slots. Two reconfiguration

operations are modeled, both with a physical cost. (i) *PA repositioning*: with pre-installed PAs that are switched on and off, a change of x_m takes microseconds to milliseconds, with sliding PAs milliseconds to seconds [11], [23], [27]; we model a change of x_m as a silence of τ_r slots of segment m . The proposed scheme changes $\mathbf{x}$ only at frame boundaries ($T_f \gg \tau_r$), whereas the reactive benchmark of Section V changes it within frames and pays τ_r at every change. (ii) *Segment activation*: the RF chain of a segment can be switched off to save energy; an active segment consumes the circuit power P_c of its RF chain in addition to its radiated power, and a segment that is switched on becomes usable only after a configuration time τ_{cfg} (a fraction of a slot for electronic activation, several slots for slower hardware). Let $\mathcal{A}[t] \subseteq \mathcal{M}$ be the set of active segments in slot t ; the energy consumed in slot t is $\mathcal{E}[t] = T_s \sum_{m \in \mathcal{A}[t]} (P_c + P_m^{\text{tx}}[t])$, and every activation $m \in \mathcal{A}[t] \setminus \mathcal{A}[t-1]$ incurs the configuration delay and, optionally, a configuration cost c_{cfg} . Whether all segments should be kept active is therefore a genuine trade-off between energy, configuration delay, and burst-draining capacity.

B. Users and Channel Model

K single-antenna HRLLC users, indexed by $k \in \mathcal{K} = \{1, \dots, K\}$, are located at $\mathbf{u}_k = [x_k, y_k, 0]^T$, uniformly distributed in $[0, D_x] \times [-D_y/2, D_y/2]$, and quasi-static within a frame; their positions are known at the BS (e.g., from the localization capability of PASS [15]). As in the PASS literature [10], [14], the link between the PA of segment m and user k is LoS dominated, and the composite channel comprises the free-space path, the in-waveguide propagation from the feed to the PA, and the in-waveguide attenuation:

$$h_{k,m}(x_m) = \frac{\sqrt{\eta} e^{-\alpha_g \ell_m}}{r_{k,m}} \exp\left(-j \frac{2\pi}{\lambda} (r_{k,m} + n_{\text{eff}} \ell_m)\right), \quad (2)$$

where $r_{k,m} = \|\mathbf{u}_k - \psi_m\| = \sqrt{(x_k - x_m)^2 + y_k^2 + d^2}$, $\ell_m = x_m - (m-1)L_s$ is the feed-to-PA distance, $\eta = c^2/(16\pi^2 f_c^2)$ with carrier frequency f_c and speed of light c , $\lambda = c/f_c$, n_{eff} is the effective refractive index of the waveguide, and $\alpha_g = \kappa \ln 10/20$ is the amplitude attenuation constant for a waveguide loss of κ dB/m. Since $\ell_m \leq L_s$, the in-waveguide loss is bounded by κL_s dB irrespective of D_x [14]. We collect the channels of user k in $\mathbf{h}_k(\mathbf{x}) = [h_{k,1}(x_1), \dots, h_{k,M}(x_M)]^T \in \mathbb{C}^M$ and define $\mathbf{H}_{\mathcal{S}}(\mathbf{x}) = [\mathbf{h}_k(\mathbf{x})]_{k \in \mathcal{S}}^H \in \mathbb{C}^{|\mathcal{S}| \times M}$ for a user subset $\mathcal{S} \subseteq \mathcal{K}$. To assess robustness, the simulations optionally add a Rician scattered component with K -factor κ_R ; the design itself uses (2).

C. Transmission Modes and Finite-Blocklength Coding

All segments share the same band of B Hz, and a slot carries $n = BT_s$ channel uses. In slot t the BS selects a subset $\mathcal{S}[t] \subseteq \mathcal{K}$ of at most M users and an active segment set $\mathcal{A}[t]$, and serves the users with linear precoding across the active feeds. Two modes arise naturally.

1) *Segment selection and aggregation (SS/SA)*: If $|\mathcal{S}| = 1$, the $j = |\mathcal{A}|$ active segments coherently serve the single user k ; $j = 1$ is segment selection (SS) and $j = M$ full aggregation. For a given j the best choice of $\mathcal{A}$ is the set $\mathcal{A}_k^{(j)}$ of the j segments with the largest $|h_{k,m}|$. Under per-segment power constraints the optimal transmission is equal-gain: every active segment transmits at power $\rho P_{\max}$ with its phase aligned to the user, where $\rho \in (0, 1]$ is a power-control factor, and the received SNR is

$$\gamma_k^{\text{SA}}(\mathbf{x}, j, \rho) = \frac{\rho P_{\max}}{\sigma^2} \left(\sum_{m \in \mathcal{A}_k^{(j)}} |h_{k,m}(x_m)| \right)^2, \quad (3)$$

where σ^2 is the noise power over B Hz. We write $\gamma_k^{\text{SA}}(\mathbf{x}, \rho)$ for the full-aggregation SNR ($j = M$).

2) *Segment multiplexing (SM)*: If $2 \leq |\mathcal{S}| \leq M$, all M segments are active and the BS applies zero-forcing (ZF) precoding $\mathbf{W}_{\mathcal{S}} = \mathbf{H}_{\mathcal{S}}^H (\mathbf{H}_{\mathcal{S}} \mathbf{H}_{\mathcal{S}}^H)^{-1}$, whose columns $\mathbf{w}_j$ satisfy $\mathbf{h}_k^H \mathbf{w}_j = \mathbb{I}\{k = j\}$, and allocates the same power p to every stream, so that segment m radiates $p \sum_{j \in \mathcal{S}} |w_{m,j}|^2$. The largest p that respects all per-segment budgets is $p(\mathcal{S}) = P_{\max} / \max_m \sum_{j \in \mathcal{S}} |w_{m,j}|^2$, and every user in $\mathcal{S}$ enjoys the interference-free SNR

$$\gamma_k^{\text{SM}}(\mathcal{S}, \mathbf{x}, \rho) = \frac{\rho p(\mathcal{S})}{\sigma^2}, \quad k \in \mathcal{S}, \quad (4)$$

with total radiated power $P_{\text{tot}}(\mathcal{S}, \rho) = \rho p(\mathcal{S}) \sum_m \sum_{j \in \mathcal{S}} |w_{m,j}|^2$. Equal-SNR ZF is chosen for its closed form: the per-slot decision then depends on the subset only through $p(\mathcal{S})$ and $P_{\text{tot}}(\mathcal{S}, 1)$, computed once per frame. An *action* of the BS in slot t is thus $\mathbf{a}[t] = (\mathcal{S}[t], \mathcal{A}[t], \mathbf{b}[t], \rho[t])$: the served users, the active segments (which fix the mode), the packets per user, and the power level.

3) *Finite-blocklength transmission*: Packets have L bits. If user $k \in \mathcal{S}$ is served with b_k packets in slot t , they are jointly encoded into one codeword of n channel uses at rate $R_k = b_k L / n$ bits per channel use. With the normal approximation [6], the decoding error probability at SNR γ is

$$\varepsilon(b, \gamma) = Q \left(\left(\log_2(1 + \gamma) - \frac{bL}{n} + \frac{\log_2 n}{2n} \right) \sqrt{\frac{n}{V(\gamma)}} \right), \quad (5)$$

with channel dispersion $V(\gamma) = (1 - (1 + \gamma)^{-2}) \log_2^2 e$. A decoding failure is detected through the cyclic redundancy check and reported over an error-free feedback channel; the b_k packets then remain in the queue and are eligible for retransmission in the next slot. Bundling more packets raises the rate and hence ε , a reliability–throughput trade-off that TAS resolves per slot and per user [8].

D. Bursty Traffic and Deadline-Constrained Queues

Packets of user k arrive according to a Markov-modulated ON–OFF process, the standard model of bursty sources [32], [35], [36]. The source alternates between an ON state, in which packets arrive as a Poisson process with h_k packets per slot (peak rate h_k / T_s), and an OFF state without arrivals; the ON→OFF and OFF→ON transition rates are α_k and β_k (in s^{-1}), so that the mean ON and OFF durations are $1/\alpha_k$

and $1/\beta_k$ and the mean arrival rate is $\bar{\lambda}_k = h_k \beta_k / (\alpha_k + \beta_k)$ packets per slot. Users may be *critical* (larger peak rate, tighter tail target) or *regular*; Section V also considers Poisson arrivals and periodic-with-jitter industrial traffic [3] with event-triggered bursts.

Each user has a first-in-first-out queue at the BS. Let $A_k[t]$ be the number of arrivals of user k in slot t and $s_k[t] \in \{0, b_k[t]\}$ the number of packets delivered (zero if user k is not scheduled or its codeword is not decoded). The delay D of a packet is the time from its arrival to its successful delivery, measured in slots and counting the transmission slot. HRLLC packets are useless after their budget: a packet whose delay would exceed $D_{\max}$ is discarded and counted as a violation, which is the deadline-dropping discipline of [5]. Denoting by $q_k[t, a]$ the number of packets of user k with age $a \in \{0, \dots, D_{\max} - 1\}$ at the beginning of slot t , the backlog is $Q_k[t] = \sum_a q_k[t, a]$ and the head-of-line (HoL) age is $\omega_k[t] = \max\{a : q_k[t, a] > 0\}$.

E. Tail-Latency Metric

The HRLLC requirement of user k is the pair $(D_{\max}, \delta_k)$: the long-run fraction of packets that are delivered later than $D_{\max}$ or discarded must not exceed δ_k ,

$$P_k^v \triangleq \lim_{T \rightarrow \infty} \frac{\sum_{t < T} (\ell_k[t] + d_k[t])}{\sum_{t < T} A_k[t]} = \Pr\{D_k > D_{\max}\} \leq \delta_k, \quad (6)$$

where $\ell_k[t]$ and $d_k[t]$ are the numbers of packets of user k delivered late and discarded in slot t , with $D_{\max} \in [0.1, 1]$ ms and $\delta_k \in [10^{-7}, 10^{-5}]$ for HRLLC [1]. Under deadline dropping, late deliveries do not occur, and P_k^v is the discard fraction; when a longer discard horizon is simulated to display the whole tail, both terms are counted. P_k^v accounts for queueing, decoding failures, and scheduling jointly, i.e., it is the end-to-end reliability [37].

III. PROBLEM FORMULATION

The BS controls two sets of variables at different rates. On the frame timescale it chooses the PA positions $\mathbf{x}[f] \in \mathcal{X}_1 \times \dots \times \mathcal{X}_M$ of frame f . On the slot timescale it chooses the action $\mathbf{a}[t] = (\mathcal{S}[t], \mathcal{A}[t], \mathbf{b}[t], \rho[t])$ with $b_k[t] \leq Q_k[t]$, $b_k[t] = 0$ for $k \notin \mathcal{S}[t]$, and $|\mathcal{S}[t]| \leq |\mathcal{A}[t]|$; the precoder follows from $\mathcal{S}[t]$, $\mathcal{A}[t]$, and $\mathbf{x}[f]$ through (3)–(4). A policy π is the collection of these decisions, and its long-run cost is $\bar{C}(\pi) \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \mathbb{E}[\mathcal{E}[t] + c_{\text{cfg}} n_{\text{on}}[t]]$ with $n_{\text{on}}[t] = |\mathcal{A}[t] \setminus \mathcal{A}[t-1]|$ the number of activations. The design objective is to meet the tail targets of all users while spending as little energy and configuration effort as possible:

$$(P1) \quad \min_{\pi} \quad \bar{C}(\pi) \quad (7a)$$

$$\text{s.t.} \quad P_k^v(\pi) \leq \delta_k, \quad \forall k \in \mathcal{K}, \quad (7b)$$

$$\mathbf{x}_m[f] \in \mathcal{X}_m, \quad \forall m, f, \quad (7c)$$

$$|\mathcal{S}[t]| \leq |\mathcal{A}[t]| \leq M, \quad \rho[t] \in (0, 1], \quad (7d)$$

$$0 \leq b_k[t] \leq Q_k[t], \text{ dynamics of Section II.} \quad (7e)$$

When the targets cannot all be met, the natural relaxation is to minimize the worst violation probability,

$\min_{\pi} \max_k P_k^v(\pi)/\delta_k$, which is the metric reported in Section V. Problem (P1) contains the rate-centric and average-delay-centric designs as special cases (no tail constraint, full activation, a rate or queue-length objective), and Section V shows that they do not solve it.

Problem (P1) is a constrained Markov decision problem with a continuous geometric decision ($\mathbf{x}$), a combinatorial decision per slot (2^K user subsets, 2^M activation sets, and $\prod_k Q_k$ packet counts), a configuration delay that couples consecutive slots, and constraints (7b) on a probability that has no closed form: the delay of a packet depends on the burst structure of all users, on the SNR tables induced by $\mathbf{x}$, on the FBL error of every bundle, and on the scheduler itself. Three observations guide the proposed solution. First, the delay tail of a single bursty source served at a constant rate is characterized in closed form by large-buffer asymptotics [29], [35], which yields a scalar rate requirement per user that captures $(h_k, \alpha_k, \beta_k, D_{\max}, \delta_k)$ jointly (Section IV-A). Second, for fixed $\mathbf{x}$ the dependence of the slot-level problem on the geometry is entirely through $O(KM + K^M)$ scalars (the SS/SA SNRs $\gamma_k^{\text{SA}}(\mathbf{x}, j, 1)$ and the SM powers $p(S)$), which can be tabulated once per frame. Third, the constraint (7b) can be handled online by a Lyapunov-type potential in the drift-plus-penalty framework [42], with the objective (7a) as the penalty, which leads to a per-slot problem that is separable over actions and users.

IV. PROPOSED TAIL-LATENCY-AWARE SCHEME: TLA-SWAN

TLA-SWAN decomposes (P1) along its two timescales. Section IV-A introduces the required-rate mapping, Section IV-B the frame-level placement (Algorithm 1), Section IV-C the slot-level scheduler (Algorithm 2), and Section IV-D the complexity and signaling. The mathematical statements are given with their proofs deferred to the appendices.

A. Tail-Latency-Aware Required Service Rate

Consider a single ON-OFF source with parameters (h, α, β) served by a constant-rate server of c packets per slot, $\bar{\lambda} < c < h$. Large-buffer asymptotics of Markov fluid queues [29], [35], [36] state that the stationary workload W satisfies $\Pr\{W > w\} \doteq e^{-\theta^*(c)w}$, where $\theta^*(c)$ is the unique positive root of $\text{EB}(\theta) = c$ and $\text{EB}(\theta)$ is the effective bandwidth of the source. Since a FIFO packet arriving to workload W experiences delay W/c , the delay-violation probability is $\Pr\{D > D_{\max}\} \doteq e^{-\theta^*(c)c D_{\max}}$. Imposing $\Pr\{D > D_{\max}\} \leq \delta$ and solving for the smallest admissible c gives the following.

Lemma 1 (Tail-latency-aware required rate). *For an ON-OFF source with peak rate h , ON-OFF rate α and OFF-OFF rate β , the effective bandwidth is $\text{EB}(\theta) = \frac{1}{2\theta} [h\theta - \alpha - \beta + \sqrt{(h\theta - \alpha - \beta)^2 + 4\beta h\theta}]$, the decay rate at service rate $c \in (\bar{\lambda}, h)$ is*

$$\theta^*(c) = \frac{(\alpha + \beta)c - \beta h}{c(h - c)}, \quad (8)$$

and the smallest constant service rate for which the large-buffer approximation of $\Pr\{D > D_{\max}\}$ does not exceed δ is

$$c_{\text{req}}(h, \alpha, \beta, D_{\max}, \delta) = h \frac{\beta + \Lambda}{\alpha + \beta + \Lambda}, \quad \Lambda \triangleq \frac{\ln(1/\delta)}{D_{\max}}. \quad (9)$$

[To do] Co-author: proof of Lemma 1 (Appendix A; item L1 of the hand-off notes).

The required rate has the expected limits: $c_{\text{req}} \rightarrow \bar{\lambda}$ when $\Lambda \rightarrow 0$ (loose target), and $c_{\text{req}} \rightarrow h$ when $\Lambda \rightarrow \infty$ (stringent target); it increases with h , with the burst duration $1/\alpha$, with $\ln(1/\delta)$, and with $1/D_{\max}$. The scalar $c_{\text{req},k} = c_{\text{req}}(h_k, \alpha_k, \beta_k, D_{\max}, \delta_k)$ is the only traffic statistic used by the placement of Section IV-B; it can be updated online from estimates of (h_k, α_k, β_k) , and Section V shows that TLA-SWAN tolerates a mismatch of a factor of two. The tail exponent $\Lambda_k = \ln(1/\delta_k)/D_{\max}$ reappears in the scheduler of Section IV-C.

B. Frame Timescale: Tail-Aware PA Placement (TAPP)

A burst of user k is drained by aggregating segments towards k , so an aggregation-mode rate is the relevant capacity for the delay tail. Because the gain of segment m at user k decays as $r_{k,m}^{-2}$, most of the aggregation gain comes from the few segments closest to the user, and using all M segments in the placement objective would pull every PA towards the single worst user and strip the other users of their nearby PAs (Section V quantifies this). We therefore measure the drain capacity of user k with its j_0 strongest segments, $j_0 \in \{1, \dots, M\}$ ($j_0 = 2$ by default):

$$R_k^{(j_0)}(\mathbf{x}) = \frac{n}{L} \left[\log_2(1 + \gamma_k^{(j_0)}) - \sqrt{\frac{V(\gamma_k^{(j_0)})}{n}} Q^{-1}(\varepsilon_0) + \frac{\log_2 n}{2n} \right], \quad (10)$$

the FBL rate of user k in packets per slot at the nominal reliability ε_0 [6], with $\gamma_k^{(j_0)} \triangleq \gamma_k^{\text{SA}}(\mathbf{x}, j_0, 1)$ from (3); $R_k^{\text{SA}} \triangleq R_k^{(M)}$ is the full-aggregation rate. The ratio $c_{\text{req},k}/R_k^{(j_0)}(\mathbf{x})$ is the fraction of slots that user k needs to be served by its j_0 best segments to drain its bursts at the tail-latency-aware rate; we call its sum over the users the *tail load*

$$\Xi(\mathbf{x}) = \sum_{k \in \mathcal{K}} \frac{c_{\text{req},k}}{R_k^{(j_0)}(\mathbf{x})}, \quad (11)$$

the time share that a round-robin server of the required rates would occupy. A tail load above one means that the users cannot all be drained at their required rates even with exclusive service, and the smaller Ξ , the more slack is left for simultaneous bursts and for multiplexing. TAPP minimizes the tail load,

$$(P2) \quad \min_{\mathbf{x} \in \mathcal{X}_1 \times \dots \times \mathcal{X}_M} \Xi(\mathbf{x}) = \sum_{k \in \mathcal{K}} \frac{c_{\text{req},k}}{R_k^{(j_0)}(\mathbf{x})}. \quad (12)$$

The max-min margin $\max_{\mathbf{x}} \min_k R_k^{(j_0)}/c_{\text{req},k}$, which protects the worst user but lets the other PAs drift, and the tail-agnostic sum-rate were also examined and found inferior (Section V). Problem (P2) is non-convex, but each coordinate x_m enters $\gamma_k^{\text{SA}}(\mathbf{x}, j_0, 1)$ only through $|h_{k,m}(x_m)| = \sqrt{\eta} e^{-\alpha_g \ell_m}/r_{k,m}$,

Algorithm 1 Tail-Aware PA Placement (TAPP), executed once per frame

Require: user positions $\{\mathbf{u}_k\}$, required rates $\{c_{\text{req},k}\}$, depth j_0 , grid size G , tolerance ξ

- 1: Initialize $x_m \leftarrow (m - \frac{1}{2})L_s$ for all m ; $\Xi \leftarrow \Xi(\mathbf{x})$ by (11)
- 2: **repeat**
- 3: $\Xi_{\text{prev}} \leftarrow \Xi$
- 4: **for** $m = 1$ **to** M **do**
- 5: Evaluate $\Xi^{(g)} = \Xi(x_1, \dots, x_m^{(g)}, \dots, x_M)$ on the grid $\{x_m^{(g)}\}_{g=1}^G \subset \mathcal{X}_m$
- 6: $g^* \leftarrow \arg \min_g \Xi^{(g)}$; **if** $\Xi^{(g^*)} < \Xi - \xi$ **then** $x_m \leftarrow x_m^{(g^*)}$, $\Xi \leftarrow \Xi^{(g^*)}$
- 7: **end for**
- 8: **until** $\Xi_{\text{prev}} - \Xi \leq \xi$

Ensure: $\mathbf{x}^* = \mathbf{x}$; if any segment moved, silence it for τ_r slots

a smooth one-dimensional function on the interval $\mathcal{X}_m$ (and through the selection of the j_0 strongest segments, which is piecewise constant). This structure suggests block-coordinate descent (BCD) with an exact one-dimensional search per block, as in Algorithm 1. The one-dimensional problem is solved by a grid of G points followed by local refinement; for a single dominant user per segment it has a closed form.

Lemma 2 (Closed-form single-user placement). *If segment m is among the j_0 strongest segments of a single user k only, the coordinate-wise minimizer of (11) is the projection of the user onto the segment corrected by the in-waveguide attenuation, $x_m^* = \Pi_{\mathcal{X}_m}(x_k - \zeta_m)$ with $\zeta_m \geq 0$ the root of a cubic equation, and $x_m^* = \Pi_{\mathcal{X}_m}(x_k)$ when $\alpha_g = 0$. If segment m serves two users k, j , x_m^* is a stationary point of $c_{\text{req},k}/R_k^{(j_0)} + c_{\text{req},j}/R_j^{(j_0)}$ on $\mathcal{X}_m$, which lies between the two projections and moves towards the user with the larger required rate.*

[To do] Co-author: proof of Lemma 2 (Appendix B; item L2).

Proposition 1 (Convergence and complexity of TAPP). *The sequence of tail loads generated by Algorithm 1 is non-increasing and bounded below by $\sum_k c_{\text{req},k}/R_k^{(j_0)}|_{r_{k,m}=d}$, hence convergent, and the algorithm terminates after a finite number of sweeps for any $\xi > 0$. Each sweep costs $O(MGK)$ evaluations of (10), and the output is a coordinate-wise optimal point of (P2) on the grid; in all our experiments the algorithm converges in two to four sweeps.*

[To do] Co-author: proof of Proposition 1 (Appendix C; item P1).

Remark 1. *The objective is a sum of inverse rates weighted by the required rates: a critical or bursty user attracts PAs even if its channel is already good, and a poorly served user is penalized much more than a well-served one is rewarded, so no user is sacrificed. The depth j_0 trades coverage against aggregation; $j_0 = 2$ was found to be a robust compromise.*

C. Slot Timescale: Tail-Aware Scheduling (TAS)

1) *Packet values and tail-risk potential:* Constraint (7b) is a long-term constraint on a tail probability and cannot be

enforced by a per-slot rule directly. Under deadline dropping, a violation is a packet whose age reaches D_{max} , so the state that matters for the tail is the age profile $\{q_k[t, a]\}$, and the natural Lyapunov-type quantity is one that grows as packets approach their deadlines. We assign to every queued packet of user k with age a the value

$$\nu_k(a) = \exp(\kappa \Lambda_k (a - D_{\text{max}})) = \delta_k^{\kappa(D_{\text{max}}-a)/D_{\text{max}}}, \quad (13)$$

with $\Lambda_k = \ln(1/\delta_k)/D_{\text{max}}$ the tail exponent of user k and $\kappa \in (0, 1]$ a tail-sharpness parameter. With $\kappa = 1$ the value of a packet is its user's violation target raised to the fraction of lifetime that remains: a fresh packet is worth δ_k , a packet at its deadline is worth one, and the value multiplies by e^{Λ_k} every slot the packet waits. The exponent Λ_k is the weight of the largest-weighted-delay-first and M-LWDF rules, for which it yields the largest possible decay rate of the delay tails of all users [39], [41]; here it is applied per packet rather than per user. The *tail-risk potential* is the total value of the queued packets,

$$\Phi[t] = \sum_{k \in \mathcal{K}} \Phi_k[t], \quad \Phi_k[t] = \sum_{a=0}^{D_{\text{max}}-1} q_k[t, a] \nu_k(a), \quad (14)$$

a weighted backlog in which a packet counts more the closer it is to expiry and the tighter the target of its user. If user k delivers its b oldest packets in slot t , with ages $a_{k,(1)} \geq a_{k,(2)} \geq \dots$, the packets that would otherwise have aged into values $\nu_k(a+1)$ leave the system, and the potential decreases by

$$U_k(b) = \sum_{i=1}^b v_k(a_{k,(i)}), \quad v_k(a) \triangleq \nu_k(a+1). \quad (15)$$

Under deadline dropping $a \leq D_{\text{max}} - 1$; when packets are kept beyond the deadline (extended discard horizon), the value is saturated at $\nu_k(D_{\text{max}}) = 1$, so that a packet that has already missed its deadline is not preferred over one that can still meet it. Since a codeword either delivers all its packets or none, the expected decrease obtained from user $k \in \mathcal{S}$ under action $a = (\mathcal{S}, \mathcal{A}, \mathbf{b}, \rho)$ is $(1 - \varepsilon(b_k, \gamma_k(a)))U_k(b_k)$, with $\gamma_k(a)$ given by (3) with $j = |\mathcal{A}|$ if $|\mathcal{S}| = 1$ and by (4) otherwise, evaluated over the segments of $\mathcal{A}$ that are already usable (segments switched on in the current slot contribute only after τ_{cfg} , or with a blocklength reduced by $\tau_{\text{cfg}}B$ when $\tau_{\text{cfg}} < T_s$). The remaining terms of the one-step drift of Φ (ageing of unserved packets, arrivals, and expiries) do not depend on the action, so that minimizing the conditional expected drift of the potential is equivalent to maximizing the expected value delivered. Adding the energy and configuration penalty of the drift-plus-penalty method [42] with parameter $V \geq 0$, and a one-step lookahead with weight $\lambda \geq 0$ that credits an action with the value $\mathcal{U}^+(a)$ it would deliver in the next slot once all its segments are usable (computed with the ages advanced by one slot), gives the per-slot problem

$$\begin{aligned} \text{(P3)} \quad & \max_{a=(\mathcal{S}, \mathcal{A}, \mathbf{b}, \rho)} \sum_{k \in \mathcal{S}} (1 - \varepsilon(b_k, \gamma_k(a)))U_k(b_k) + \lambda \mathcal{U}^+(a) \\ & - V \left[\sum_{m \in \mathcal{A}} (P_c + P_m^{\text{tx}}(a))T_s + c_{\text{cfg}} |\mathcal{A} \setminus \mathcal{A}[t-1]| \right], \quad (16) \end{aligned}$$

subject to (7d). The lookahead lets the controller switch on segments that cannot help immediately; with $\lambda = 0$, $P_c = 0$, and $c_{\text{cfg}} = 0$, (P3) is the pure tail-risk minimization. In the language of risk-aware control [2], (P3) is the Lagrangian form of “choose the cheapest action whose risk score $\mathbb{E}[\Phi[t+1] | a]$ is acceptable”.

2) *Solution of the per-slot problem.* For fixed $(\mathcal{S}, \mathcal{A}, \rho)$, (P3) separates over users, and the optimal bundle size of user k is

$$b_k^*(\mathcal{S}, \mathcal{A}, \rho) = \arg \max_{1 \leq b \leq \min\{Q_k, b_{\max}\}} (1 - \varepsilon(b, \gamma_k(a))) U_k(b), \quad (17)$$

a one-dimensional integer search with at most $b_{\max}$ candidates, where $b_{\max}$ is the largest number of packets that fits in a slot at the maximum SNR. Because $U_k(b)$ is increasing and concave in b while $1 - \varepsilon$ decreases in b , (17) bundles many packets when they are young and sends only the urgent ones, at a lower and more reliable rate, when the oldest packets approach their deadline; this is the tail-aware counterpart of queue-aware FBL coding [8]. The outer maximization over $(\mathcal{S}, \mathcal{A}, \rho)$ is exhaustive over a structured action set: the SS/SA actions $(\{k\}, \mathcal{A}_k^{(j)})$ for every user and every depth $j \in \{1, \dots, M\}$, and the SM actions $(\mathcal{S}, \mathcal{M})$ for every subset with $2 \leq |\mathcal{S}| \leq M$, i.e., $KM + \sum_{2 \leq s \leq M} \binom{K}{s}$ actions (186 for $K = 8$, $M = 4$). Their SNRs and the error terms $\varepsilon(b, \rho\gamma)$ for all $b \leq b_{\max}$ and all ρ in a finite set $\mathcal{R}$ are tabulated once per frame, so that evaluating (P3) is a vectorized lookup, feasible for $K \leq 12$ and $M \leq 8$ within a fraction of a slot (Section IV-D); for larger K the search can be restricted to the users with the largest HoL values or built greedily. The mode is a by-product of (P3) rather than a separate rule: with energy cost, the controller uses SS when a single user holds moderately urgent packets, deepens the aggregation as the urgency grows, and multiplexes when several users hold packets of comparable urgency; without energy cost, deeper aggregation is never worse and the controller behaves as a pure tail minimizer. Algorithm 2 summarizes the procedure.

Remark 2 (Relation to virtual-queue designs). A virtual credit that accumulates whenever a backlogged user is served below $c_{\text{req},k}$ [43], shifting the ages in (15) by $\zeta Z_k / c_{\text{req},k}$, was also implemented; because $c_{\text{req},k}$ is close to the peak rate for HRLLC targets, the credit over-prioritizes the bursting user and degrades the tail monotonically with ζ (from 1.2×10^{-3} at $\zeta = 0$ to 2.5×10^{-3} at $\zeta = 1$ in the default scenario), so the packet-value formulation is used without credit.

3) *Guarantees.* The following statements formalize the behavior of Algorithm 2.

Theorem 1 (Drift-plus-penalty characterization). Let $\Delta(\Phi[t]) = \mathbb{E}[\Phi[t+1] - \Phi[t] | \{q_k[t, \cdot]\}]$ denote the conditional one-step drift of the tail-risk potential (14). Then (i) $\Delta(\Phi[t]) + V \mathbb{E}[P_{\text{tot}}[t] | \{q_k[t, \cdot]\}] = C[t] - \sum_{k \in \mathcal{S}[t]} e^{\kappa \Lambda_k} (1 - \varepsilon_k[t]) U_k(b_k[t]) + V P_{\text{tot}}[t]$, where $C[t]$ does not depend on the action, so that Algorithm 2 with the user weights $e^{\kappa \Lambda_k}$ absorbed in $v_k(\cdot)$ minimizes the drift-plus-penalty expression exactly in every slot; (ii) if the arrivals are bounded, $A_k[t] \leq A_{\max}$,

Algorithm 2 Tail-Aware Scheduling (TAS), executed every slot t

Require: action tables from $\mathbf{x}^*$ (SNR, power, segments, $\varepsilon(b, \rho\gamma)$ for $b \leq b_{\max}$, $\rho \in \mathcal{R}$); age profiles $\{q_k[t, \cdot]\}$; active set $\mathcal{A}[t-1]$, readiness times; $\kappa, V, \lambda, P_c, c_{\text{cfg}}, \tau_{\text{cfg}}$

- 1: Admit new arrivals with age 0; discard packets with age $\geq D_{\max}$ and count them as violations
- 2: Compute $v_k(a)$ by (15) and $U_k(b)$ for $b \leq \min\{Q_k, b_{\max}\}$, $\forall k$ with $Q_k > 0$ (and with the ages advanced by one slot if $\lambda > 0$)
- 3: **for** every action $(\mathcal{S}, \mathcal{A})$ with backlogged users and every $\rho \in \mathcal{R}$ **do**
- 4: SNR over the usable segments of $\mathcal{A}$; b_k^* by (17) for $k \in \mathcal{S}$; $J(a)$ by (16)
- 5: **end for**
- 6: $a^* \leftarrow \arg \max J(a)$; switch on the segments of $\mathcal{A}^* \setminus \mathcal{A}[t-1]$ (usable after τ_{cfg}) and off the others; transmit b_k^* packets to each $k \in \mathcal{S}^*$ with EGT if $|\mathcal{S}^*| = 1$ and ZF otherwise
- 7: On ACK, remove the b_k^* oldest packets of k ; on NACK keep them; age all remaining packets by one slot; if no user is backlogged, keep the active set powered (keep-warm) or switch all segments off

and there exists a stationary randomized policy π and $\epsilon > 0$ such that $\mathbb{E}_\pi[\text{value delivered in a slot} | \Phi] \geq (e^{\kappa \Lambda_{\max}} - 1)\Phi / e^{\kappa \Lambda_{\max}} + \epsilon \Phi$ whenever Φ exceeds a threshold, then the time-average potential under Algorithm 2 with $V = 0$ is bounded, $\bar{\Phi} \leq \mathcal{B}_\Phi / \epsilon$, and the delay-violation probability of user k satisfies $P_k^v \leq e^{\kappa \Lambda_k} \bar{\Phi}_k / \bar{\lambda}_k$, where $\bar{\Phi}_k$ is the time-average of Φ_k ; (iii) for $V > 0$ the time-average power is within $O(1/V)$ of that of π at the price of an $O(V)$ increase of $\bar{\Phi}$.

[To do] Co-author: proof of Theorem 1 (Appendix D; item T1).

Theorem 2 (Tail bound). Let $\mu_k(\theta)$ denote the effective capacity [31], [38] of the service process $\{s_k[t]\}$ of user k induced by TLA-SWAN, and let $\text{EB}_k(\theta)$ be the effective bandwidth of its ON-OFF source. If $\theta_k^* > 0$ solves $\text{EB}_k(\theta) = \mu_k(\theta)$, then $\Pr\{D_k > D_{\max}\} \leq \varsigma_k e^{-\theta_k^* \text{EB}_k(\theta_k^*) D_{\max}}$ with $\varsigma_k = \Pr\{Q_k > 0\} \leq 1$. In particular, if the fraction of slots in which user k is served in SA mode is at least ϕ_k and the SA bundle is decoded with probability at least $1 - \epsilon_0$, then $\mu_k(\theta) \geq -\frac{1}{\theta} \ln(1 - \phi_k + \phi_k(\epsilon_0 + (1 - \epsilon_0)e^{-\theta b_k^{\text{SA}}}))$, where $b_k^{\text{SA}} = \lfloor R_k^{\text{SA}}(\mathbf{x}^*) \rfloor$, which makes the bound computable from the output of Algorithm 1.

[To do] Co-author: proof of Theorem 2 (Appendix E; item T2).

D. Complexity and Signaling

TAPP runs once per frame with $O(IMGK)$ operations ($I \leq 5$ sweeps in all our experiments). The per-frame tables cost $O(KM + \sum_{2 \leq s \leq M} \binom{K}{s} s^3)$ plus $|\mathcal{R}|b_{\max}$ error values per action, i.e., 186 actions and about 7×10^3 scalars for $K = 8$, $M = 4$. Per slot, TAS evaluates (15) in $O(KD_{\max})$ and (P3) in $O(|\mathcal{R}|b_{\max}K(KM + \sum_s \binom{K}{s}))$ vectorized operations, a few 10^4 multiply-adds and one $\arg \max$. Signaling is that of

TABLE I
DEFAULT SIMULATION PARAMETERS

Parameter	Value
f_c ; n_{eff} ; waveguide loss κ	28 GHz; 1.4; 0.08 dB/m
M , L_s ; d ; D_y	4, 10 m; 3 m; 10 m
Noise σ^2 ($B = 2$ MHz, NF 10 dB); P_{max}	-101 dBm; -10 dBm
K ; L ; T_s ; n ; b_{max}	8; 256 bits; 0.1 ms; 200; 12
h ; mean ON/OFF durations	3 pkt/slot; 1 ms / 9 ms
D_{max} ; δ ; ε_0 ; critical	1 ms; 10^{-5} ; 10^{-5} ; $h = 4$, $\delta = 10^{-6}$
τ_r ; τ_{cfg} ; j_0 ; κ ; λ	1 slot; 0; 2; 0.75; 0 (1 if $\tau_{\text{cfg}} > 0$)
Energy-aware runs: P_c ; V ; $\mathcal{R}$	100 mW; 0.3; {0.25, 0.5, 1}

a multi-user MIMO downlink (CSI, ACK/NACK) plus three traffic statistics per user updated per frame.

V. SIMULATION RESULTS

A. Setup and Benchmarks

We simulate the system of Section II at the slot level, including the FBL decoding errors drawn from (5), retransmissions, deadline dropping, the reconfiguration silences, and the exact packet ages, with the parameters of Table I unless stated otherwise; the PHY parameters follow [14], [17], the 0.1-ms slot with $n = 200$ channel uses corresponds to a 2-MHz carrier, and the 32-byte packets to the 3GPP reference payload [3]. The default ON-OFF traffic has a 1-ms mean burst, an activity factor of 0.1 (per-slot transition probabilities $1 - e^{-\alpha T_s}$, $1 - e^{-\beta T_s}$), and a peak rate $h = 3$ packets per slot, i.e., a mean load of 2.4 packets per slot for $K = 8$ users against a single-stream capacity of 3 to 5 packets per slot: a single burst is drainable, simultaneous bursts overload the system. Each point pools three or four user drops of 1.5×10^5 slots, i.e., at least 10^6 packets, so that violation probabilities down to 10^{-5} are resolved; points with no observed violation are drawn at $1/(\text{number of packets})$.

The benchmarks are: **M-LWDF** [39], **max-weight** (queue-length weights, average-delay oriented), and **rate-max** (expected delivered packets, queue-blind) control on the same action set and TAPP placement; **EDF** (the M users with the oldest HoL packets, channel-agnostic); the rate-centric **sum-rate placement with max-weight** [14], [16]; **fixed PAs** at the segment centers; **fixed SS** and **fixed SA** (one segment, respectively all segments, one user per slot); **reactive repositioning** (every slot each segment moves its PA above the backlogged user with the largest value and pays τ_r slots of silence); the **conventional PASS** with a single waveguide (one RF chain, M PAs at the TAPP positions, in-waveguide loss over D_x , SA/TDMA only) [10]; and a **fixed array** of M co-located antennas at the center of the area with ZF and TAS.

B. Tail Versus Average Delay (H1)

Fig. 2 tests H1: for the same six controllers and the same packets (kept until 4 ms so that the mean is well defined) it shows the mean delay (Fig. 2(a)) and the probability of exceeding the 1-ms budget (Fig. 2(b)). The mean delays of the four channel-aware SWAN controllers are indistinguishable up to $h = 3$ (0.10–0.16 ms), and the throughput-greedy rate-max and max-weight rules even have the smallest mean at high

load, but their tails differ by orders of magnitude: at $h = 2$, TLA-SWAN and M-LWDF deliver every one of 9.6×10^5 packets within the budget, whereas max-weight and rate-max miss it with probability 1.1×10^{-4} and 1.7×10^{-4} , and at $h = 2.5$ the four controllers span 3.9×10^{-4} (TLA-SWAN), 5.9×10^{-4} (M-LWDF), 1.3×10^{-3} (max-weight), and 1.8×10^{-3} (rate-max). A design that reports only the mean delay would declare max-weight or rate-max the best scheduler and miss the HRLLC target by two orders of magnitude. Fixed full aggregation (TDMA) and EDF are worse in both statistics, and all controllers converge when the load approaches saturation ($h \geq 3.5$). Fig. 3 shows the whole delay tail at $h = 3$ (4×10^6 packets per curve): the conventional PASS, the fixed array, and EDF miss the budget for 9–11% of the packets, the rate-centric design and max-weight for 1.3×10^{-2} and 7.1×10^{-3} , fixed PAs for 4.5×10^{-3} , and the tail-aware controllers for 3.4×10^{-3} (M-LWDF) and 2.7×10^{-3} (TLA-SWAN), with mean delays of all SWAN controllers between 0.15 and 0.18 ms. Beyond 2 ms the M-LWDF curve falls below that of TLA-SWAN: once a packet has missed its deadline its value (13) saturates and TLA-SWAN no longer prioritizes it over packets that can still be delivered in time; under the deadline-dropping discipline of the remaining figures this residual lateness does not exist. The 99.99th and 99.999th delay percentiles of TLA-SWAN are 2.8 and 3.8 ms against 3.7 and 4.1 ms for max-weight.

C. Traffic Burstiness and Latency Budget

Fig. 4 varies the traffic and the requirement under deadline dropping (1.1×10^6 packets per point). In Fig. 4(a) the peak rate grows at a fixed activity factor: at $h = 2$ TLA-SWAN discards no packet, M-LWDF 1.1×10^{-5} , fixed PAs 3.7×10^{-5} , max-weight 1.4×10^{-4} , the rate-centric design 4.3×10^{-4} , and the conventional PASS 3.4×10^{-3} ; the ordering is preserved over the whole range, with TLA-SWAN 1.3–1.7 times below M-LWDF and 2.5–6 times below max-weight up to $h = 3$, and the curves converge only when the peak rate approaches the capacity of four streams ($h = 4$). Fig. 4(b) keeps the mean rate fixed and varies the mean burst duration from 0.25 to 4 ms: burstiness alone moves the tail of TLA-SWAN by four orders of magnitude, from 9×10^{-7} to 8×10^{-3} , and the advantage of tail-aware control is largest for short and medium bursts (a factor 500 over max-weight at 0.25 ms, 4 at 1 ms) and shrinks to 1.5 for 4-ms bursts, where the system is simply overloaded during a burst and only more segments help. Fig. 4(c) varies the budget from 0.3 to 3 ms: with 3 slots every scheme discards about 2% of the packets, the budget then buys about one order of magnitude per doubling for the tail-aware controllers, which reach 5×10^{-5} (TLA-SWAN) and 1.4×10^{-4} (M-LWDF) at 2 ms and no discarded packet, respectively 1.8×10^{-5} , at 3 ms, whereas max-weight and the rate-centric design remain at 3.3×10^{-4} and 8.3×10^{-4} . The same trend holds versus the number of users: with $K = 4$ TLA-SWAN discards no packet against 1.7×10^{-4} for max-weight, and with $K = 12$ (three users per segment) the two reach 8.3×10^{-3} and 2×10^{-2} . Tail-aware control is most valuable where the resources are sufficient on average but not during simultaneous bursts.

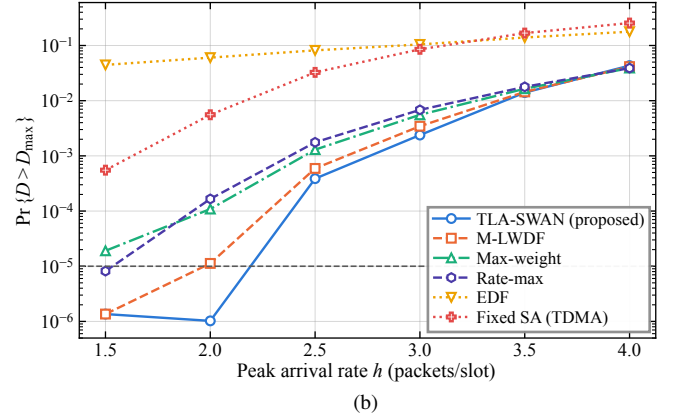
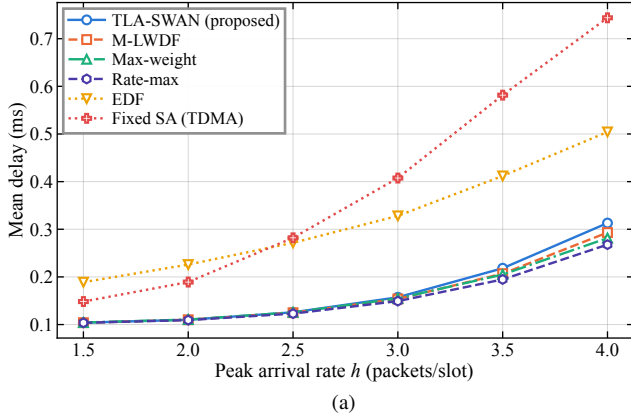


Fig. 2. (a) Mean delay and (b) delay-violation probability of the same controllers versus the peak arrival rate; packets are kept until 4 ms so that both statistics are measured on the same packets (dashed line: target $\delta = 10^{-5}$).

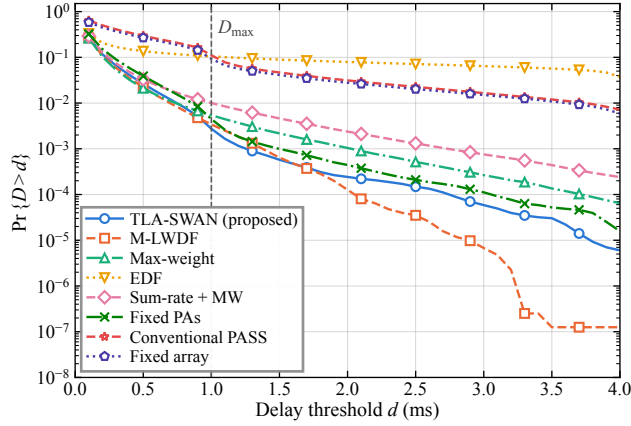


Fig. 3. Complementary CDF of the packet delay for the default parameters (discard horizon 4 ms).

D. SWAN Resources and PA Repositioning

Fig. 5(a) varies the per-segment power, i.e., the SNR of a user 5 m from its nearest PA from 6 to 26 dB. Below -15 dBm every scheme discards more than 4% of the packets because a stream carries one or two packets per slot; the tail then falls by about an order of magnitude per 5 dB, and at -5 dBm TLA-SWAN discards 5×10^{-6} (6 of 1.1×10^6), M-LWDF 4×10^{-5} , max-weight 4×10^{-4} , and the fixed array 5.5×10^{-3} ; at 0 dBm TLA-SWAN discards none while max-weight still discards 3.6×10^{-5} . The gain of tail-aware control grows with the SNR, from 4 at -10 dBm to 70 at -5 dBm over max-weight, and the fixed array needs about 10 dB more power than the SWAN for the same tail. Fig. 5(b) divides the 40-m area into M segments: two segments cannot serve eight bursty users ($> 8\%$), four give 1.2×10^{-3} , five 1.1×10^{-4} , and eight leave no discarded packet in 1.1×10^6 for the tail-aware controllers, whereas the conventional PASS with the same PAs improves only from 0.16 to 5×10^{-3} : segmentation, i.e., multiplexing plus short feed-to-PA distances, is worth two to three orders of magnitude at equal number of PAs and is the resource that removes the overload regime of Fig. 4(b). Fig. 5(c) contrasts TAPP with reactive repositioning: even with

instantaneous moves the reactive policy is 1.6 times worse (1.8×10^{-3} against 1.1×10^{-3}), because chasing the most urgent user takes the PA away from the users whose bursts come next, and it degrades to 3.4×10^{-2} for $\tau_r = 10$ slots with 6×10^3 moves per 10^5 slots, whereas TAPP is unaffected. PA repositioning is thus best treated as a slow, statistics-driven operation.

E. Segment Activation, Energy, and Configuration Delay (H2)

Fig. 6 addresses H2 with a circuit power of $P_c = 100$ mW per active RF chain. In Fig. 6(a) the energy weight V trades power against the tail: without energy awareness every controller keeps 1.5 segments active on average (150 mW) and TLA-SWAN attains 9.3×10^{-4} against 4.4×10^{-3} (max-weight) and 3.4×10^{-3} (rate-max); with $V = 0.3$ TLA-SWAN needs 94 mW for 4.4×10^{-3} , the tail that max-weight reaches at 138–150 mW, and at 92–97 mW the tail-agnostic controllers, driven to the same power by a larger V , are five to six times worse (2.2 – 2.6×10^{-2}). For equal reliability, tail-aware control saves about 35% of the energy, and for equal energy it improves the tail by a factor of 5 (H3). Fig. 6(b) shows the mechanism at $V = 0.3$: segment selection is used almost exclusively while the backlog is below ten packets, aggregation deepens and multiplexing starts as it grows, and multiplexing is used in two thirds of the slots above 40 packets; the segments are activated exactly when bursts need them. Fig. 6(c) quantifies the cost of hyper-reliability at $h = 2$, where the targets are reachable, as the minimum power over V at which a target is met: TLA-SWAN meets 10^{-3} , 10^{-4} , and 10^{-5} with 62, 63, and 66 mW and discards none of 1.3×10^6 packets with 85 mW, so that tightening the target by three orders of magnitude costs about 40% more energy, whereas max-weight and rate-max need 90 and 105 mW for 10^{-3} and never reach 10^{-5} , since even with all segments permanently active they discard about 10^{-4} . Fig. 6(d) varies the activation delay for the energy-aware controller ($\lambda = 1$): sub-slot delays (electronic activation) raise the tail from 2.5×10^{-3} to 4.0×10^{-3} and 6.1×10^{-3} at 0.25 and 0.5 slot, delays of one, two, and four slots give 8.2×10^{-3} , 2×10^{-2} , and 8×10^{-2} , keeping the last active set powered

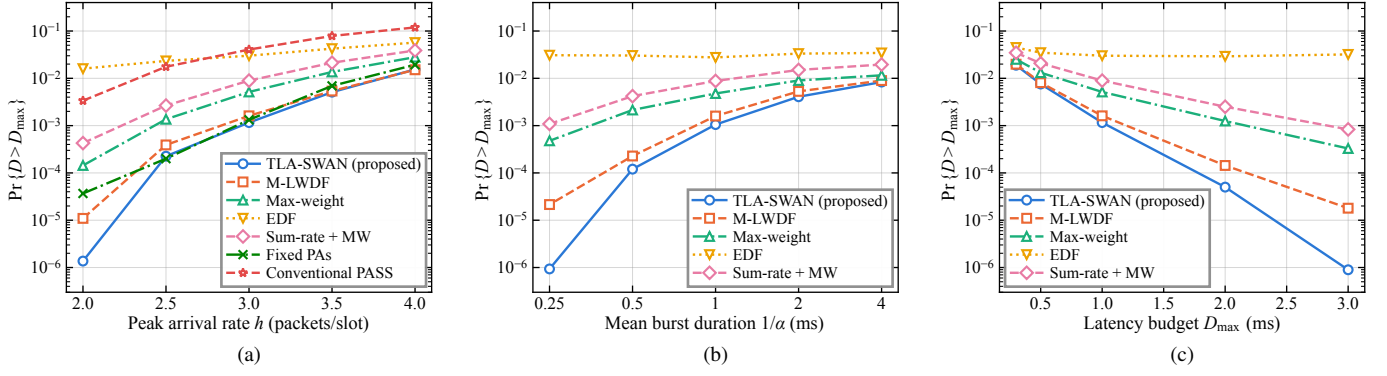


Fig. 4. Delay-violation probability under deadline dropping versus (a) the peak arrival rate h , (b) the mean burst duration $1/\alpha$ at a fixed mean rate, and (c) the latency budget $D_{\max}$.

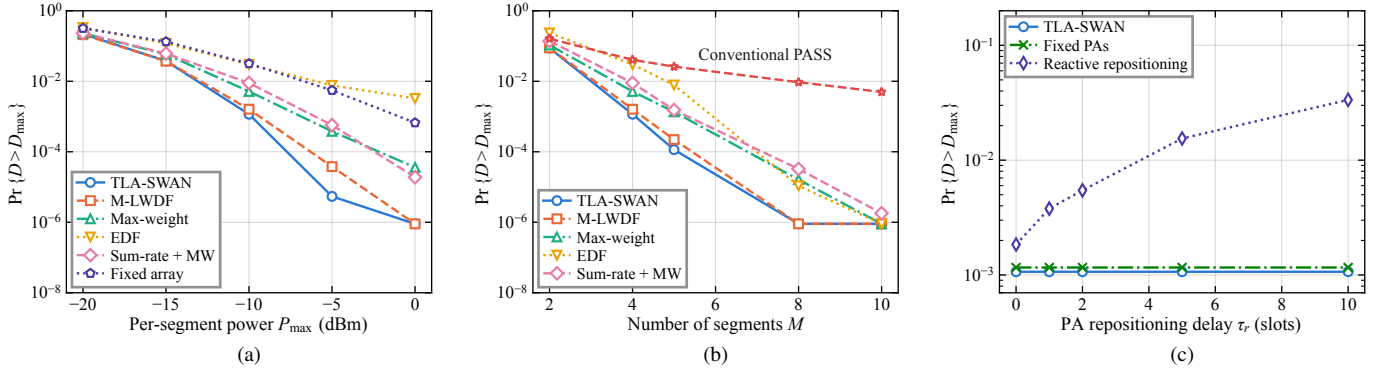


Fig. 5. Delay-violation probability versus (a) the per-segment transmit power $P_{\max}$, (b) the number of segments M for a 40-m service area, and (c) the PA repositioning delay τ_r (frame-level TAPP placement, fixed PAs at the segment centers, and per-slot reactive repositioning).

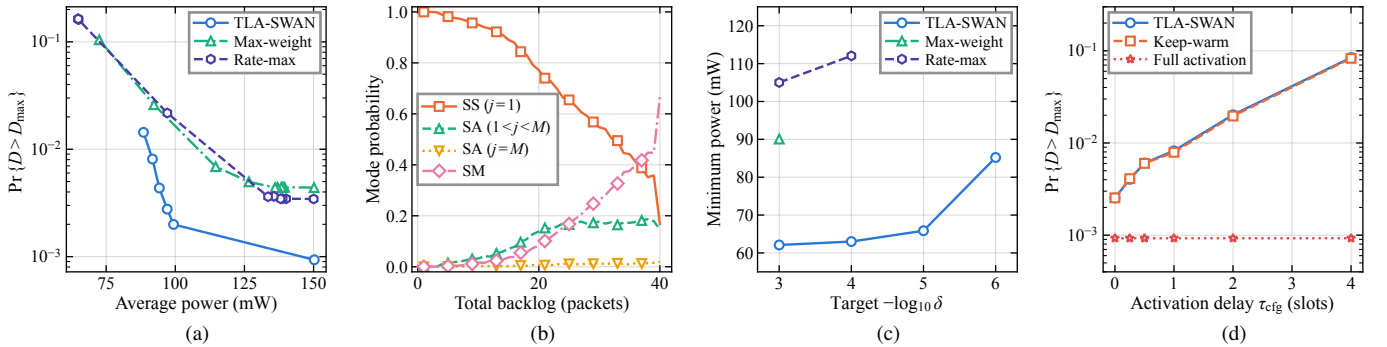


Fig. 6. Energy-aware operation with $P_c = 100$ mW per active RF chain: (a) average power versus delay-violation probability, one marker per energy weight V (increasing from right to left); (b) probability of each SWAN mode of TLA-SWAN ($V = 0.3$) versus the total backlog; (c) minimum average power at which each controller meets a target δ at $h = 2$ (missing points: target not reachable); (d) delay-violation probability versus the segment activation delay τ_{cfg} .

while idle does not help (40 mW more, same tail), and keeping all segments permanently active is immune (9.3×10^{-4}) at four times the energy. Full segment activation is therefore not the energy-optimal operating point of a SWAN under bursty traffic unless activation takes several slots, in which case the choice is again a trade-off between energy and the tail.

F. Arrival Process (H4) and Analytical Validation

Fig. 7(a) isolates the segment dimension with a TDMA controller of fixed aggregation depth j under ON-OFF and Poisson arrivals of equal mean: under Poisson arrivals $j = 2$ removes all but one violation in 1.4×10^6 packets, so an

energy-aware policy would activate two segments, whereas under ON-OFF arrivals the discard rate keeps decreasing with j (from 0.135 to 0.031 at $j = M$) and TDMA is in any case insufficient, which is why the proposed controller multiplexes during bursts. With Poisson arrivals of the default mean, none of the channel-aware controllers discards a single packet of 1.1×10^6 , whereas the ON-OFF and the periodic-with-jitter industrial model (a 2-packet batch every 2 ms plus event bursts) separate tail-aware and tail-agnostic controllers by factors of 4 to 7 (2.4×10^{-3} against 1.0×10^{-2} for the latter): the burst structure, not the mean load, is what a SWAN HRLLC controller must be designed against (H4). Figs. 7(b)

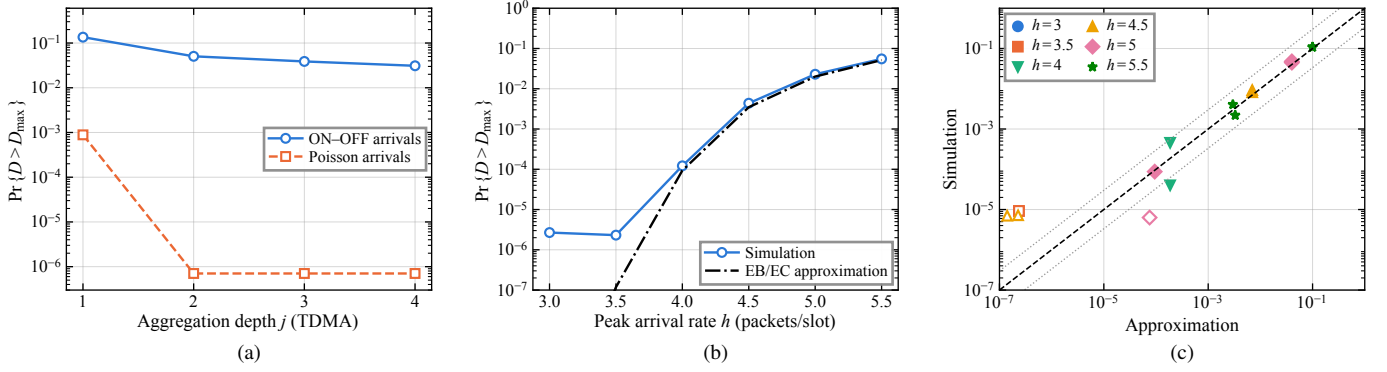


Fig. 7. (a) TDMA control with a fixed aggregation depth j under ON-OFF and Poisson arrivals of equal mean rate; (b) single device served in full aggregation ($P_{\max} = -15$ dBm): simulation versus the effective-bandwidth (EB/EC) approximation; (c) drop-by-drop comparison of (b) (open markers: no observed violation; dotted lines: a factor of three).

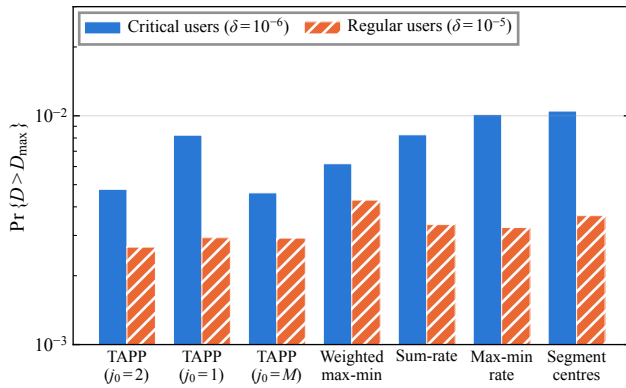


Fig. 8. Per-class delay-violation probability for seven placement rules with two critical ($h = 4$, $\delta = 10^{-6}$) and six regular users over 16 drops (TAS scheduling).

TABLE II
ABLATIONS OF THE ENERGY-AWARE TLA-SWAN ($\tau_{\text{cfg}} = 0.5$ SLOT, $\lambda = 1$, $V = 0.3$)

Variant	$\Pr\{D > D_{\max}\}$	Power (mW)
Energy-aware TLA-SWAN (reference)	6.1×10^{-3}	112
Shannon rate instead of FBL	3.9×10^{-3}	108
No in-waveguide loss	6.4×10^{-3}	108
Poisson arrivals (same mean)	$< 10^{-6}$	164
No activation delay	2.5×10^{-3}	97
Full activation (all segments on)	1.1×10^{-3}	233
Queue-blind rate-max, same V	6.6×10^{-3}	143

and 7(c) validate the analytical route on the single-device case of Theorem 2: one ON-OFF device served every slot in full-aggregation mode has an i.i.d. FBL service ($b^* = 4$ or 5 packets per slot depending on the drop), and the closed-form effective-bandwidth approximation follows the simulation over four orders of magnitude, is within a factor of 2.5 of every drop with observed violations, and predicts below 10^{-7} whenever none is observed in 7×10^5 packets.

G. Placement, Robustness, and Ablations

Fig. 8 examines the placement with two critical users among eight over 16 drops, because the effect of the placement depends strongly on the geometry. TAPP ($j_0 = 2$) gives the

lowest violation probability, 3.3×10^{-3} overall and 4.8×10^{-3} for the critical users, against 4.9×10^{-3} and 8.3×10^{-3} for the sum-rate placement, 5.4×10^{-3} and 1.0×10^{-2} for the tail-agnostic max-min placement, and 5.8×10^{-3} and 1.0×10^{-2} for PAs at the segment centers: weighting the placement by the required rates halves the violation probability of the critical users at no cost for the regular users. Among the variants, $j_0 = 1$ is 1.4 times worse, $j_0 = M$ marginally worse, and the weighted max-min margin 1.5 times worse overall although it yields the smallest worst-user probability. Algorithm 1 converges in two to four sweeps in every drop. The design is robust: with a Rician scattered component of 20, 10, and 4.8 dB, the tail of the LoS-designed TLA-SWAN grows from 9.7×10^{-4} to 1.0, 1.4, and 3.6×10^{-3} while max-weight grows from 3.8×10^{-3} to 9.5×10^{-3} ; a two-fold mismatch of the critical users' required rates changes the overall violation probability by less than 5% and only a four-fold underestimation raises their probability from 4.8×10^{-3} to 8.3×10^{-3} ; and the tail varies by $\pm 25\%$ over $\kappa \in [0.25, 1.5]$. Table II removes one ingredient at a time from the energy-aware controller: FBL coding costs a factor 1.6 with respect to a Shannon-rate model, the in-waveguide loss of 10-m segments is immaterial, the activation delay costs a factor 2.4, full activation buys a factor 5.7 at twice the energy, and a queue-blind rate-maximizing controller with the same energy penalty is both worse in the tail and 28% more power hungry.

VI. CONCLUSIONS

This paper has studied, for the first time, a SWAN that serves HRLLC users with bursty, deadline-constrained traffic and finite-blocklength codes, and has taken the delay tail rather than the rate as the design target. The tail-latency-aware required rate summarizes the burst statistics and the tail target of a user in one closed-form scalar; minimizing the tail load moves the antennas towards the users whose bursts are hardest to drain; and a packet-value controller that minimizes the drift of a tail-risk potential plus an energy penalty selects, slot by slot, the users, the active segments, the packets per codeword, and the power. Slot-level simulations showed that controllers with equal average delays differ by orders of magnitude in the tail, that TLA-SWAN improves the delay-

violation probability by factors of 2 to 70 over rate- and delay-oriented control and by orders of magnitude over conventional PASS and fixed arrays, that it meets a 10^{-5} target with 35% less energy, that full segment activation is not energy-optimal, that bursty arrivals change the optimal segment policy, and that the effective-bandwidth approximation predicts the single-device tail. Future work includes segment activation under a limited number of RF chains, uplink and grant-free access with tail guarantees, and correlated bursts across users.

APPENDIX A

PROOFS OF LEMMAS 1 AND 2, PROPOSITION 1, AND THEOREMS 1 AND 2

[To do] Co-author: Appendices A–E (items L1, L2, P1, T1, T2 of docs/math_todo_for_colleague.md).

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